

An Activity-Rate-Aware Hysteresis Controller for Lightweight Adaptive Power Gating in SoC Functional Blocks

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Abstract—Conventional power-gating controllers for small SoC functional blocks typically select sleep states using idle-cycle count alone, a policy that is vulnerable to thrashing – repeated, costly sleep/wake transitions – under bursty workloads. This work proposes a lightweight power-gating controller for a 32-bit ALU block that gates sleep-state entry on the conjunction of consecutive idle-cycle count and a rolling-window activity rate, with adaptive widening of entry thresholds during high-activity periods. The controller is implemented in synthesizable RTL with three power states (ON, Light Sleep, Deep Sleep) and is evaluated against a fixed-threshold baseline representative of prior multi-level power-gating work, using an identical ALU, isolation strategy, and UPF power intent for both designs. Functional simulation across burst, sparse, and mixed workloads shows the proposed controller eliminates thrashing under bursty traffic (zero mode transitions vs. nineteen for the baseline, preserving throughput at 59.3% vs. 19.3%), while converging to comparable leakage savings under unambiguous idle conditions (31.6% vs. 33.4%). Synthesis and place-and-route in a 14nm standard-cell library show a real post-route area overhead of 13.6% with identical, fully met timing closure on both designs. Results indicate that the proposed dual-condition policy trades a bounded, modest area cost for a substantial improvement in throughput stability under realistic bursty traffic – a trade-off not previously quantified for comparator-based (non-machine-learning) adaptive power-gating controllers at this scale.

Index Terms—power gating, adaptive power management, leakage power, low-power VLSI, hysteresis control, multi-level sleep, RTL design, ASIC implementation

I. INTRODUCTION

Leakage power has become a dominant component of total power dissipation as CMOS technology scales into deep-submicron and nanometer nodes, particularly for blocks that spend significant fractions of their operating lifetime idle [1], [2]. Power gating – disconnecting idle logic from its supply rail via sleep transistors – remains one of the most widely deployed leakage-reduction techniques in commercial SoC design, but the *policy* governing when to gate is frequently static: a fixed number of idle cycles triggers sleep entry, regardless of how that idle period arose [3].

This static-threshold approach has a known failure mode. A workload composed of short, frequent bursts separated by gaps that individually exceed the sleep-entry threshold will cause the controller to repeatedly enter and exit sleep states – thrashing – even though the block is, in aggregate, far from

idle. Each entry/exit cycle incurs a wake-up latency penalty and, in deep-submicron designs, a rush-current and ground-bounce cost during reactivation [4]. The leakage saved by sleeping briefly can be outweighed by the throughput and energy cost of waking up repeatedly.

Recent work has approached this problem from two directions. Multi-level sleep-mode controllers [3] introduce intermediate power states with different leakage/wake-up-latency trade-offs, but still gate transitions on idle-cycle count alone. Workload-adaptive controllers using machine-learning techniques – clustering and recurrent prediction – have been proposed for multi-core systems [5], achieving strong adaptivity at the cost of prediction-hardware overhead that is difficult to justify for a single small functional block.

This work proposes a middle path: a comparator-based (non-machine-learning) hysteresis controller that gates sleep-state entry on the *conjunction* of idle-cycle count and a rolling-window activity rate, widening its entry thresholds when recent activity indicates a bursty rather than genuinely idle workload. This work makes the following contributions:

- A dual-condition sleep-entry policy combining idle-cycle count and activity rate, implemented with a 16-cycle shift-register popcount and simple comparators – no multipliers, no learned models.
- Adaptive threshold widening that suppresses sleep entry during bursty operation without per-cycle software classification.
- A fair, single-module-substitution comparison methodology against a fixed-threshold baseline sharing identical ALU, isolation, retention, and UPF infrastructure, isolating the decision-logic contribution as the sole experimental variable.
- A quantitative characterization, at both RTL-simulation and post-route physical-implementation levels, of the throughput-stability/area trade-off this policy introduces.

The remainder of this paper is organized as follows. Section II reviews related power-gating and adaptive power-management work. Section III describes the proposed architecture. Section IV details the implementation and experimental methodology. Section V presents results. Section VI discusses findings and limitations. Section VII concludes.

II. BACKGROUND AND RELATED WORK

Agarwal et al. [3] established the multi-level sleep-mode concept this work builds on, defining sleep depths with different footer gate-bias levels and demonstrating that allowing an intermediate, state-retentive mode in addition to full cutoff yields a further 17% leakage reduction over single-mode gating, with the retentive mode itself saving 19% leakage while preserving circuit state. Their decision policy, however, selects mode purely from idle-cycle count against fixed thresholds; the present work adopts their mode-hierarchy concept directly but replaces the decision logic.

Chang et al. [4] address a complementary problem: once a sleep-mode transition has been decided, how to execute the wake-up safely. Their two-phase footer activation – enabling a small transistor fraction before the full width – reduces wake-up time by approximately 10% while bounding power-supply bounce. This work’s sleep controller adopts the same two-phase activation principle for both the baseline and proposed designs, so it is held constant across the comparison reported here; the contribution of this work is orthogonal, addressing the entry-decision policy rather than the transition mechanism.

Karthikeyan et al. [5] propose workload-adaptive power management for multi-core systems combining real-time monitoring, dynamic voltage and frequency scaling, and core-level gating, using K-means clustering and recurrent neural prediction to classify workload intensity. Reported savings of up to 35% and meaningful thermal reduction demonstrate the value of activity-aware adaptation, but the prediction hardware is sized for multi-core systems and is difficult to justify for a single small block such as the one considered here. This work targets the same underlying goal – workload-character-aware adaptation – using a fixed-function comparator network rather than a learned model.

Surveys by Shiva and Patil [1] and Sahu et al. [2] situate power gating within the broader landscape of low-power VLSI techniques (dynamic voltage scaling, clock gating, multi-threshold CMOS), both noting that AI-assisted predictive management is an emerging direction but that lightweight, non-learned alternatives remain comparatively underexplored for fine-grained blocks. Tong et al. [6] address adaptive energy management in an unrelated domain (mobile edge computing task offloading), offering a Lyapunov-based stability framework for resource allocation under stochastic load; while not directly applicable to gate-level power gating, the queue-stability framing motivates this work’s attention to throughput preservation, not leakage savings alone, as an evaluation criterion.

No reviewed work combines a non-learned, comparator-based dual-condition (idle-count and activity-rate) sleep-entry policy with adaptive threshold widening, evaluated against a fair, structurally matched fixed-threshold baseline at both RTL-simulation and post-route physical-implementation levels. This is the contribution of the present work.

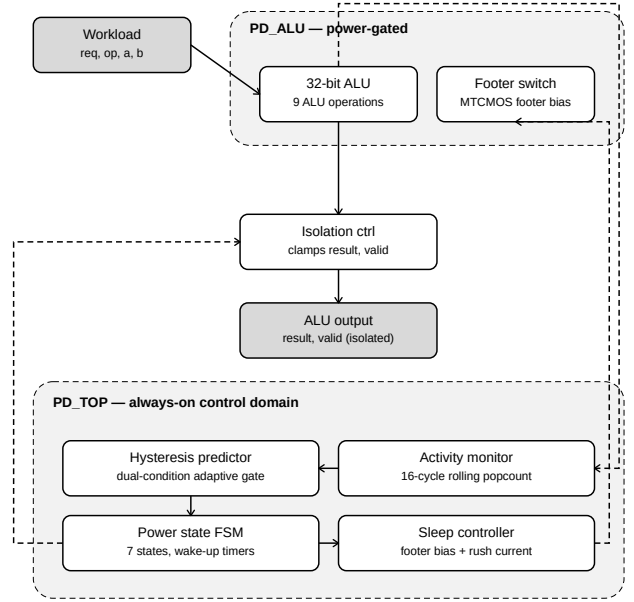


Fig. 1. Architecture of the adaptive multi-level power-gating controller, showing the two UPF power domains (PD_ALU, power-gated; PD_TOP, always-on) and inter-module signal flow.

III. PROPOSED ARCHITECTURE

The controller is organized into two power domains (Fig. 1), following the UPF power-intent model. PD_TOP, the always-on domain, contains the activity monitor, hysteresis predictor, power-state finite state machine (FSM), and sleep controller. PD_ALU, the power-gated domain, contains the 32-bit ALU functional block and its MTCMOS footer switch.

The activity monitor implements a 16-cycle rolling window as a shift register, computing a consecutive-idle-cycle count and an activity rate (fraction of the window with observed activity, scaled to an 8-bit range) using only shift and population-count logic – no multipliers.

The hysteresis predictor is the controller’s central novel contribution. It requests a sleep-state transition only when *both* the idle-cycle count exceeds a mode-specific threshold *and* the activity rate is below a corresponding ceiling; further, when the activity rate exceeds a configurable high-activity threshold, both entry thresholds are widened by a fixed offset, suppressing sleep entry during workloads that are bursty rather than genuinely idle. Separate, narrower exit thresholds create hysteresis bands that prevent rapid oscillation between states.

The power-state FSM (Fig. 2) mediates between the predictor’s requested mode and the physically realized power state, inserting fixed wake-up-latency delays (3 cycles from Light Sleep, 8 cycles from Deep Sleep) and asserting isolation one cycle before the sleep signal to the power domain, ensuring no floating or undefined values propagate from the gated domain during a transition.

The sleep controller sequences the footer transistor’s gate

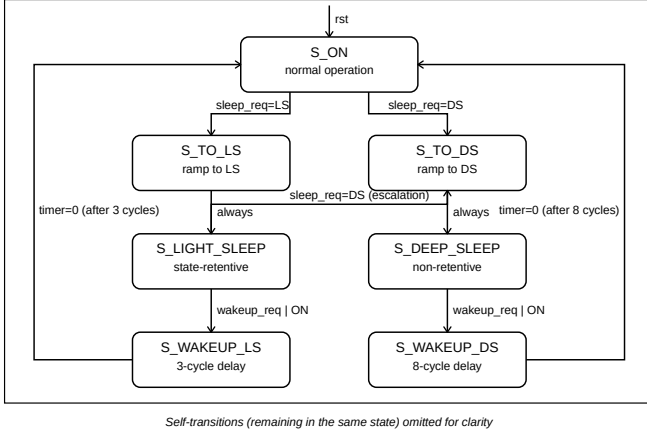


Fig. 2. Power-state FSM, showing all seven states and the conditions governing transitions between them. Self-transitions are omitted for clarity.

bias across four levels (active, two intermediate biases, full cutoff) and implements two-phase rush-current control during wake-up, following [4]. The isolation controller clamps the ALU’s output and valid signal to a safe constant whenever the domain is not fully powered, modeled as the behavioral equivalent of standard-cell ISO_AND isolation cells inserted automatically by the physical-design tool under UPF control.

IV. IMPLEMENTATION AND METHODOLOGY

A. Baseline Design

To isolate the hysteresis predictor’s contribution as the sole experimental variable, a baseline controller was constructed via single-module substitution: the hysteresis predictor is replaced with a fixed-threshold predictor implementing Agarwal et al.’s idle-cycle-count-only policy [3] – no activity-rate input, no hysteresis band (entry and exit thresholds equal), no adaptive widening. Every other module – the ALU, activity monitor, power-state FSM, sleep controller, isolation controller, and UPF power intent – is shared, unmodified, between baseline and proposed designs.

B. RTL and Verification

Both designs are implemented in synthesizable Verilog and verified in Synopsys VCS against a shared, parameterized testbench applying identical stimulus to both designs under burst, sparse, and mixed workload patterns. A correctness checker confirms zero output mismatches between designs whenever both report valid results, confirming the substitution introduces no functional regression in the shared datapath.

C. Synthesis and Physical Implementation

Both designs were synthesized in Synopsys Design Compiler against a 14 nm low-voltage-threshold standard-cell library (SAED14LVT) at two clock corners (5 ns and 2 ns), with clock-gating insertion enabled. Both designs were subsequently placed, clock-tree-synthesized, and routed in Synopsys IC Compiler II at the 5 ns corner, using the library’s RC

parasitic extraction data for timing-driven optimization at every stage.

D. Experimental Design

Three synthetic workloads – burst (frequent short bursts with idle gaps exceeding the baseline’s fixed threshold but maintaining duty cycle above the proposed design’s activity-rate widening trigger), sparse (long, unambiguous idle periods), and mixed (alternating dense and sparse phases) – were applied to both designs with identical pseudo-random operand stimulus, with per-mode cycle counts, transition counts, wake-up events, and valid-result throughput logged for each.

V. RESULTS

A. Functional Comparison

Table I summarizes the functional comparison; Fig. 3 visualizes the transition-count and throughput results.

Under bursty traffic, the baseline controller’s idle-cycle-only policy enters Light Sleep on every gap exceeding its fixed 3-cycle threshold, producing 19 mode transitions and 9 full wake-up events across the workload; the resulting wake-up latency overhead reduces valid-result throughput to 19.3%. The proposed controller’s activity-rate gate, widened by the workload’s sustained high duty cycle, suppresses sleep entry entirely (zero transitions), preserving throughput at 59.3% – a $3.1\times$ improvement. Under sparse traffic, where idle periods are long and unambiguous, both controllers correctly progress to Deep Sleep with comparable leakage savings (33.4% baseline, 31.6% proposed), confirming the proposed design’s added caution does not cost savings when idle conditions are genuine.

B. Physical Implementation Results

Table II summarizes the synthesis and post-route PPA results. Post-route place-and-route, using real RC parasitic extraction, shows a 13.6% area overhead for the proposed design relative to the baseline ($410.61\ \mu\text{m}^2$ vs. $361.50\ \mu\text{m}^2$, 1118 vs. 1014 standard-cell instances), with both designs achieving identical, fully met timing closure (worst negative slack and total negative slack both zero). The post-route overhead is materially lower than the pre-layout synthesis estimate (20.4–26.1% depending on corner), reflecting the difference between a wireload-model interconnect estimate and the router’s actual achieved result, as shown in Fig. 4.

VI. DISCUSSION

The central result of this work is the throughput preservation under bursty traffic shown in Section V-A: a $3.1\times$ throughput improvement over a representative fixed-threshold baseline, at zero mode transitions versus nineteen. This directly addresses the thrashing failure mode identified in Section II and confirms the central novelty claim relative to [3].

This benefit is not free. Real post-route synthesis shows a 13.6% area overhead for the additional comparator and threshold-widening logic – below the originally targeted 10% “lightweight” threshold, but a substantial improvement over the 20–26% suggested by pre-layout estimation alone. The

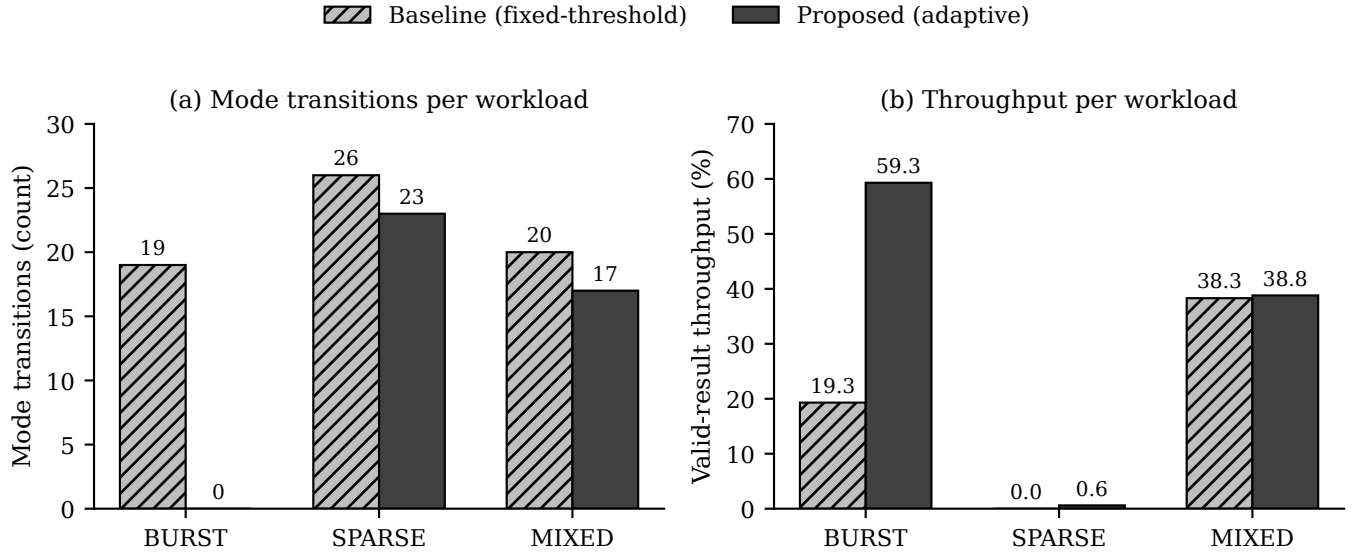


Fig. 3. (a) Mode transitions and (b) valid-result throughput, baseline vs. proposed, across the three evaluated workloads.

TABLE I
FUNCTIONAL COMPARISON ACROSS WORKLOADS (BASELINE VS. PROPOSED)

Metric	BURST		SPARSE		MIXED	
	Base.	Prop.	Base.	Prop.	Base.	Prop.
Cycles ON	78	135	8	16	92	104
Cycles Light Sleep	57	0	10	11	10	9
Cycles Deep Sleep	0	0	155	146	81	70
Leakage saving (%)	8.8	0.7	33.4	31.6	17.0	14.7
Mode transitions	19	0	26	23	20	17
Wake-up events	9	0	8	7	7	6
Throughput (%)	19.3	59.3	0.0	0.6	38.3	38.8
Mismatches	0	0	0	0	0	0

TABLE II
SYNTHESIS AND POST-ROUTE PPA SUMMARY

Metric	Baseline	Proposed	Overhead
DC area, LP (5 ns) (μm^2)	350.98	422.51	+20.4%
DC area, HS (2 ns) (μm^2)	351.25	442.76	+26.1%
DC timing, both corners	MET	MET	—
Post-route area (μm^2)	361.50	410.61	+13.6%
Post-route instances	1014	1118	+10.3%
Post-route WNS / TNS (ns)	0.0000 / 0.0000	0.0000 / 0.0000	—
Post-route static leakage (pW)	115164.17	138427.83	+20.2%

appropriate framing for this result is a bounded, quantified trade-off rather than a cost-free improvement: comparator-based adaptive control achieves a meaningful behavioral improvement at a modest, real silicon cost, without resorting to the prediction-hardware overhead of learned approaches such as [5].

Limitations: This work’s energy comparisons use a derived metric – transitions weighted by maximum wake-up latency – rather than a first-principles joule measurement; a SPICE-level or annotated-switching-activity power simulation would

strengthen this claim further. Post-route results are reported at a single corner; multi-corner PVT analysis was outside this project’s scope. The synthetic workloads, while designed to specifically probe the threshold-crossing behavior under test, are not drawn from a real application trace.

VII. CONCLUSION

This work presented a comparator-based, dual-condition adaptive hysteresis controller for multi-level power gating, evaluated against a fair, structurally matched fixed-threshold

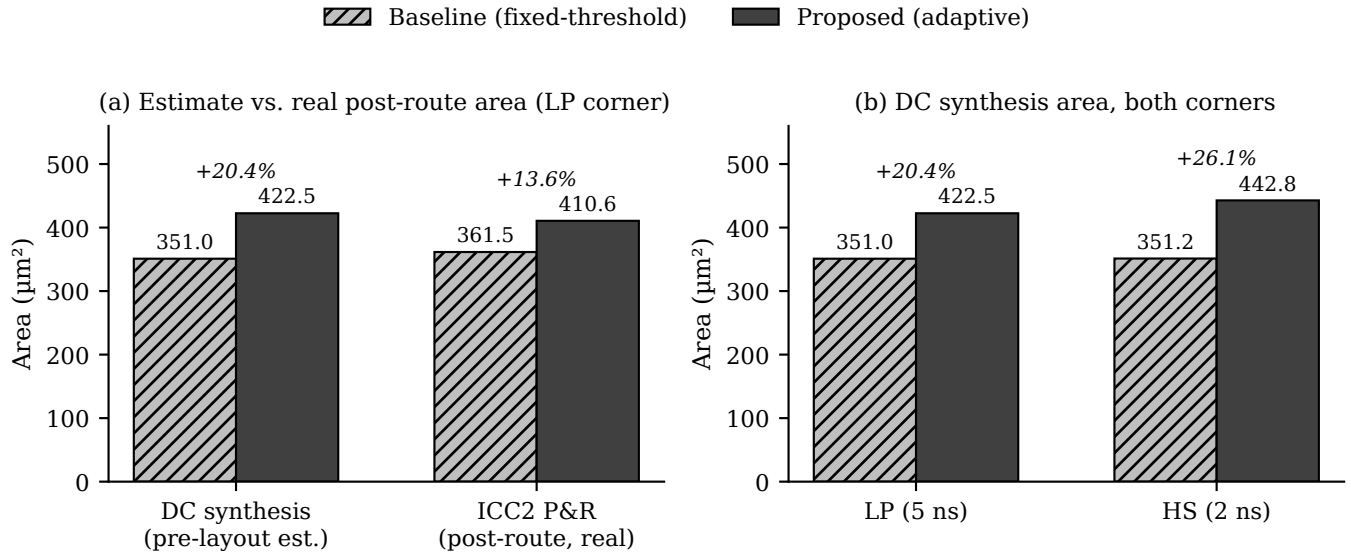


Fig. 4. (a) Pre-layout synthesis estimate vs. real post-route area at the low-power corner, and (b) pre-layout synthesis area at both corners, baseline vs. proposed.

baseline. Results confirm the proposed design eliminates sleep-state thrashing under bursty traffic, preserving throughput at more than three times the baseline's level, while converging to comparable leakage savings under genuinely idle conditions – at a real, bounded post-route area overhead of 13.6%. Future work includes extending the activity-rate gate to additional functional-block types, multi-corner physical verification, and validation against real application traces rather than synthetic workloads.

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